

CHARTERHALL, United Kingdom

WMO: 031580

Lat: 55.7086N

Lon: 2.3847W

Elev: 111

StdP: 100.00

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.2	-5.6	2.4	-0.8	-4.5	2.6	0.0	16.4	8.4	14.8	7.8	2.4	290	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	23.2	17.6	21.1	16.6	19.7	15.6	18.7	21.9	17.4	20.2	16.4	18.8	4.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.7	19.9	16.2	11.7	18.7	15.3	11.0	17.6	53.3	22.0	49.4	20.4	46.3	18.9	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.2	9.8	-6.5	26.6	2.4	2.1	-8.2	28.1	-9.6	29.4	-10.9	30.6	-12.7	32.1		
(5)			-6.1	21.4	1.7	2.3	-7.3	23.0	-8.3	24.4	-9.3	25.6	-10.5	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	3.5	4.1	5.5	7.4	9.8	12.6	14.7	14.6	12.6	9.4	6.0	3.7
	DBStd	4.82	2.74	2.80	2.69	2.65	2.66	2.28	2.38	2.22	2.16	2.68	2.87	3.34
	HDD10.0	995	202	167	141	84	36	3	0	0	3	42	123	194
	HDD18.3	3527	460	399	398	327	264	172	116	119	173	276	369	452
	CDD10.0	517	0	1	2	7	30	81	146	142	80	25	3	1
	CDD18.3	6	0	0	0	0	0	3	2	1	0	0	0	0
	CDH23.3	56	0	0	0	0	0	3	29	22	2	0	0	0
	CDH26.7	6	0	0	0	0	0	0	2	3	0	0	0	0
Wind	WSAvg	4.4	4.9	5.3	5.1	4.5	4.2	3.8	3.4	3.7	4.1	4.4	4.4	4.7
Precipitation	PrecAvg	863	81	61	55	60	60	68	71	76	66	88	93	84
	PrecMax	1122	183	132	113	144	104	166	152	162	132	183	171	180
	PrecMin	643	25	14	23	14	17	16	28	15	22	28	32	33
	PrecStd	123	34	30	22	35	24	38	35	37	30	38	35	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.3	15.0	18.3	21.1	23.4	26.4	26.3	22.2	17.6	14.4	12.4
		MCWB	9.9	10.1	10.7	12.4	15.7	17.4	18.5	19.4	17.3	14.6	12.5	11.1
	2%	DB	10.1	10.7	12.5	15.2	18.6	20.7	23.6	22.4	19.8	15.9	12.7	10.9
		MCWB	8.7	9.0	9.1	11.1	14.3	16.2	18.0	17.5	15.7	13.7	11.2	9.7
	5%	DB	8.9	9.5	11.0	13.4	16.3	19.0	21.4	20.4	18.0	14.7	11.5	9.9
		MCWB	7.6	7.9	8.5	10.0	12.5	14.9	17.1	16.3	14.6	12.8	10.1	8.7
	10%	DB	7.9	8.5	9.8	12.0	14.7	17.3	19.5	18.9	16.7	13.6	10.5	8.7
		MCWB	6.6	7.0	7.7	9.1	11.4	13.8	15.9	15.3	13.9	11.9	9.2	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	10.6	11.4	13.4	17.4	18.9	20.9	20.3	18.6	15.6	12.7	11.3
		MCDB	11.3	12.0	13.8	17.0	20.0	21.8	23.1	22.6	21.1	16.7	13.9	12.2
	2%	WB	9.0	9.2	9.9	11.6	14.6	16.9	18.9	18.2	16.5	14.1	11.5	10.0
		MCDB	10.0	10.5	11.8	14.5	17.7	19.7	22.7	21.3	18.5	15.3	12.5	10.9
	5%	WB	7.8	8.2	8.9	10.6	13.1	15.5	17.5	17.0	15.3	13.1	10.4	8.9
		MCDB	8.8	9.4	10.6	12.9	15.7	18.2	20.5	19.8	17.3	14.5	11.3	9.9
	10%	WB	6.8	7.2	7.9	9.5	11.8	14.3	16.4	16.0	14.2	12.1	9.4	7.7
		MCDB	7.8	8.4	9.7	11.7	14.0	16.7	19.0	18.2	16.0	13.4	10.3	8.7
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	6.5	7.3	7.9	7.9	8.6	8.6	7.7	6.4	5.5	4.8
		MCDBR	5.8	6.1	8.1	9.6	10.7	10.9	11.8	10.6	10.1	7.2	6.2	6.0
		MCWBR	5.2	5.0	5.7	6.2	6.9	7.0	7.1	6.3	6.6	5.4	5.5	5.6
	5% WB	MCDBR	5.7	5.8	7.5	9.1	9.8	10.0	10.5	9.7	9.1	6.8	6.0	6.0
		MCWBR	5.2	5.1	5.8	6.3	7.0	7.1	7.2	6.6	6.9	5.7	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.313	0.329	0.359	0.380	0.378	0.379	0.379	0.369	0.353	0.336	0.311	0.295	
	taud	2.286	2.375	2.319	2.289	2.341	2.363	2.367	2.407	2.460	2.455	2.354	2.252	
	Ebn at Noon	594	742	802	838	863	865	856	843	808	727	598	517	
	Edh at Noon	55	72	96	114	115	114	111	100	83	66	51	44	
All-Sky Solar Radiation	RadAvg	0.53	1.17	2.22	3.62	4.76	4.63	4.46	3.61	2.58	1.41	0.69	0.39	
	RadStd	0.08	0.12	0.25	0.39	0.33	0.43	0.45	0.27	0.20	0.16	0.06	0.04	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page